



UNIVERSITY OF
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University
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Acoustic Signal Classification

Final Report

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Abstract

This report presents the findings of a project focused on classifying acoustic signals from mine roofs to assess their safety. The classification is based on audio recordings labeled as “Drummy” or “Tight”, indicating the condition of the mine roof. The project utilized machine learning techniques to analyze and classify these audio signals, contributing to improved safety measures in mining operations. The results demonstrate the effectiveness of the developed models in accurately identifying roof conditions based on acoustic data.

1 Background and Project Overview

Evaluating whether an impact sounds tight or drummy is a subjective skill that requires training and experience. Challenging conditions (e.g., intersecting rooms, high roofs) can make this judgment difficult even for experts. This project records the sound of scaling-bar impacts and uses machine learning to classify them as either “Tight” or “Drummy”, reducing subjectivity. **Phase 2**

Device Functionality:

- Detect impacts and record audio.
- Optional in-field labeling (drummy/tight).
- Transmit audio via Bluetooth to the app.
- Receive classifier results and display via RGB LED.

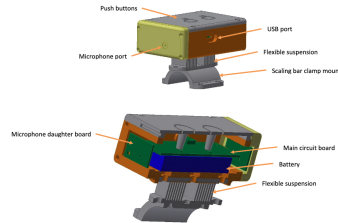


Figure 1: Phase 2 custom device mounted on a scaling bar.

2 Understanding the Dataset

2.1 Phase 1

Phase 1 consists of **3,309 recordings** (**2,212 drummy** and **1,097 tight**). Each recording is **0.75 s** long (**36,000 samples**) at **48 kHz**. The class distribution is imbalanced (drummy $\approx 2\times$ tight), so any accuracy $\leq 66\%$ is not better than a majority guess.

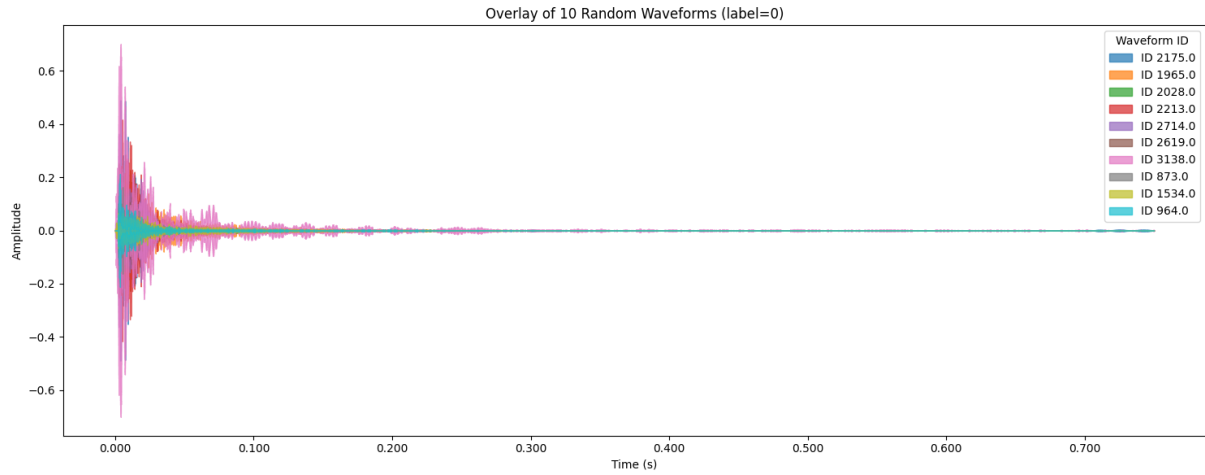


Figure 2: Waveform overlay of Phase 1 recordings — Label 0 (Drummy).

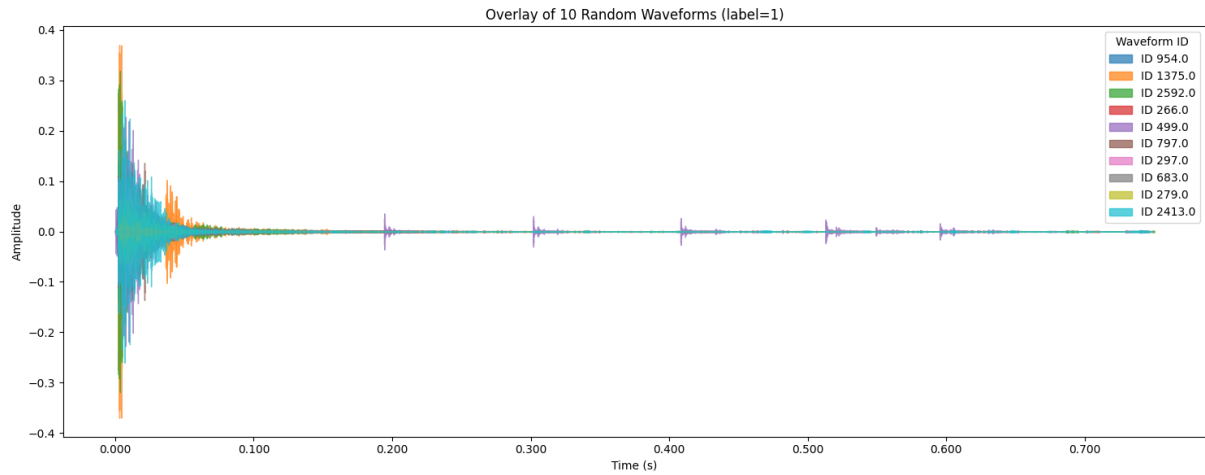


Figure 3: Waveform overlay of Phase 1 recordings — Label 1 (Tight).

2.2 Phase 2

Raw Phase 2 contains **9,923** recordings; **1,483** were **NaN-corrupted** ($\Rightarrow$ 8,440). In addition, **mishits** (non-impacts/invalid recordings) were identified and removed by waveform screening, yielding a final dataset of **8,229** recordings. Each has **4,800 samples** (**0.1 s** at **48 kHz**). Classes are more balanced; any accuracy $\leq$ **54%** is not better than a majority guess.

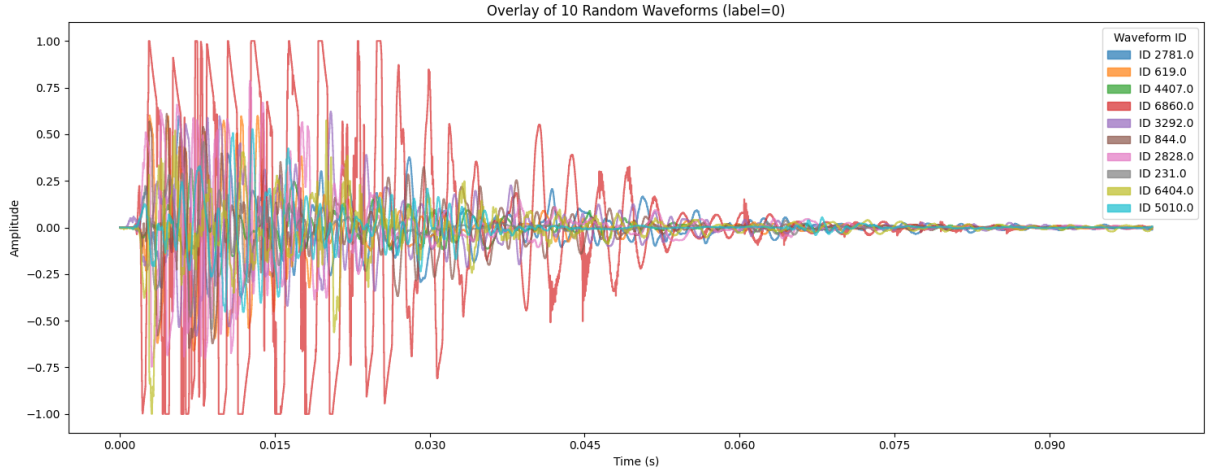


Figure 4: Waveform overlay of Phase 2 recordings — Label 0 (Drummy).

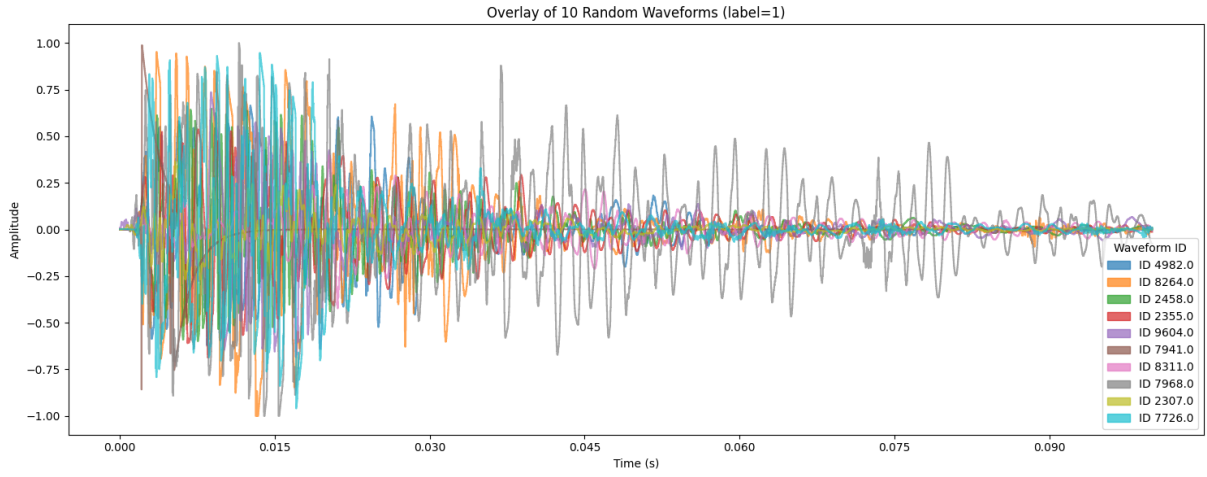


Figure 5: Waveform overlay of Phase 2 recordings — Label 1 (Tight).

2.3 Mishits

Mishits were identified by visual inspection of the waveform and spectrogram.

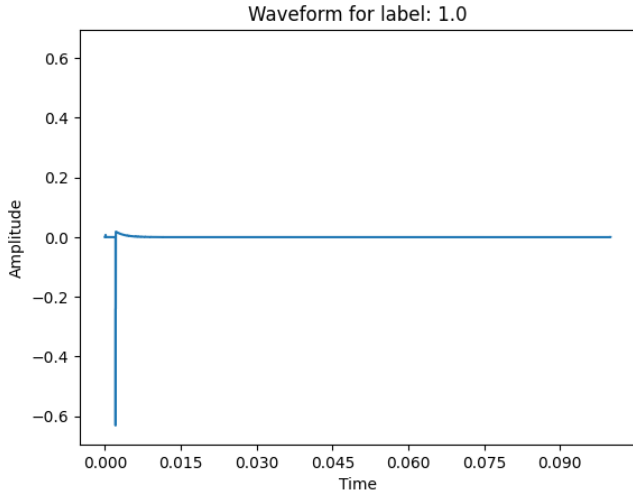


Figure 6: Phase 2 mishit (Tight).

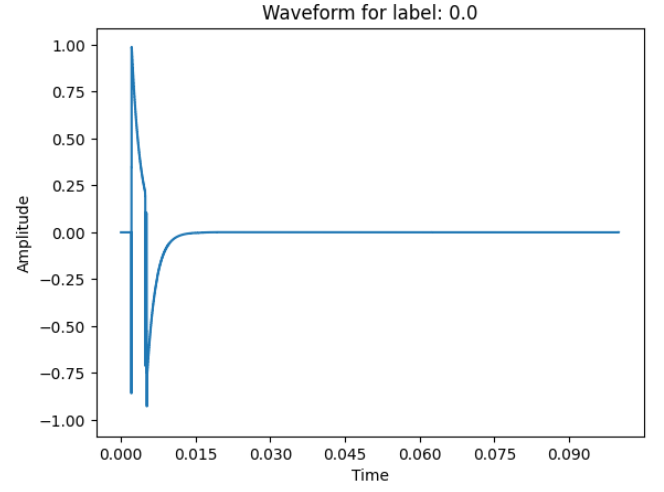


Figure 7: Phase 2 mishit (Drummy).

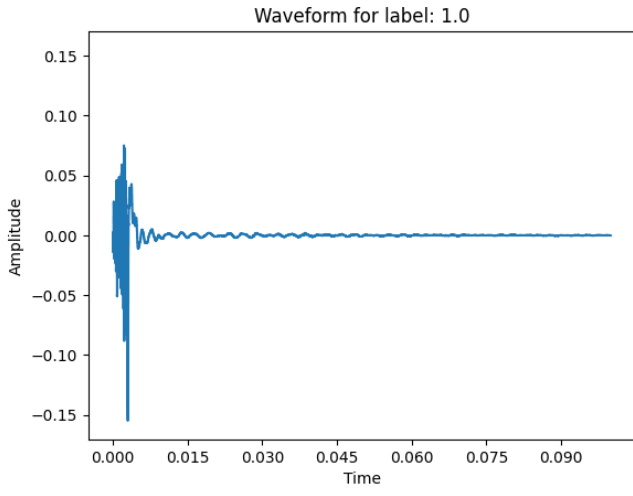


Figure 8: Phase 2 mishit (Tight).

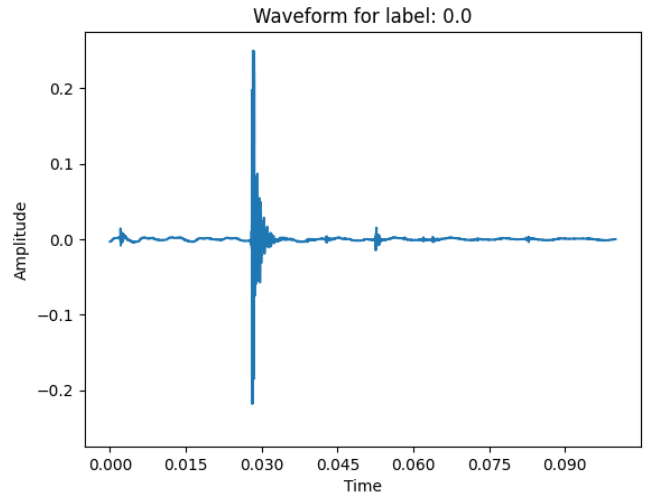


Figure 9: Phase 2 mishit (Drummy).

3 PCA as Feature Extraction

PCA was used for dimensionality reduction; explained variance guided feature count. We compared SVM and MLP; **MLP consistently outperformed SVM**. Split: **80% train / 20% test**.

3.1 Phase 1

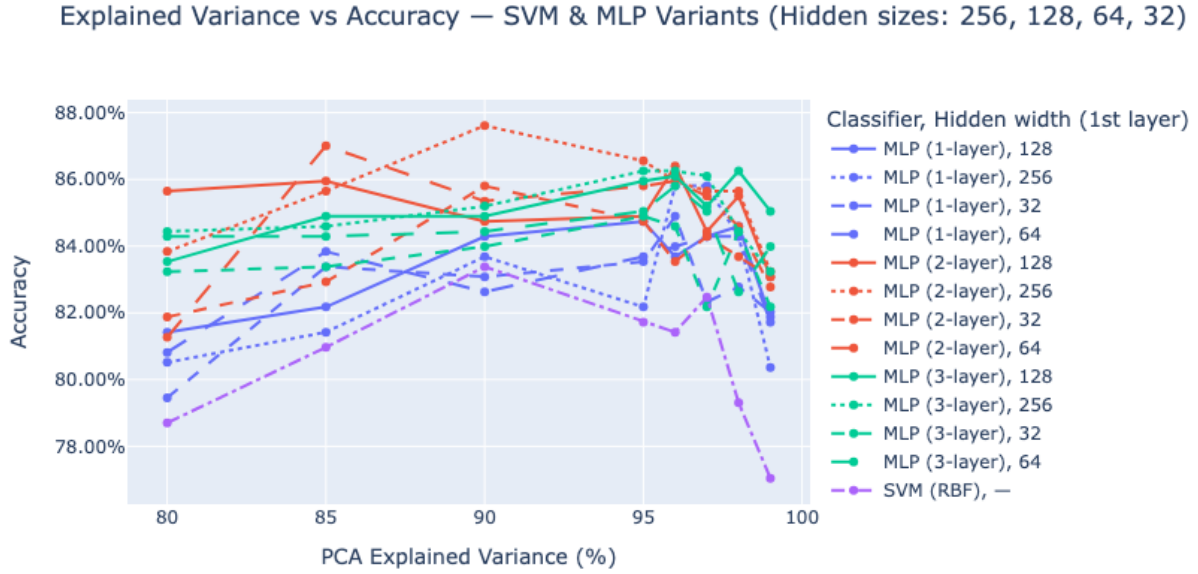


Figure 10: Phase 1 — PCA explained variance vs model accuracy (multiple models).

Best Model and Feature Combination:

90% explained variance, MLP with 2 hidden layers (256, 128), dropout 0.2.

Table 1: Phase 1 PCA — Best model classification report.

	precision	recall	f1-score	support
drummy (0)	0.88	0.91	0.89	443
tight (1)	0.80	0.74	0.77	219
accuracy		0.85		662
macro avg	0.84	0.83	0.83	662
weighted avg	0.85	0.85	0.85	662

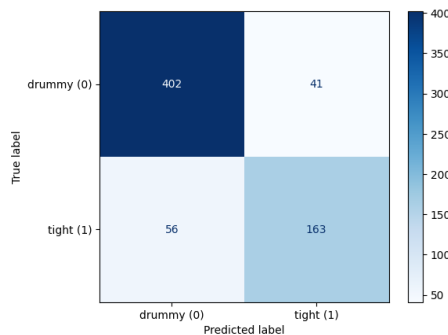


Figure 11: Phase 1 PCA — Best model confusion matrix.

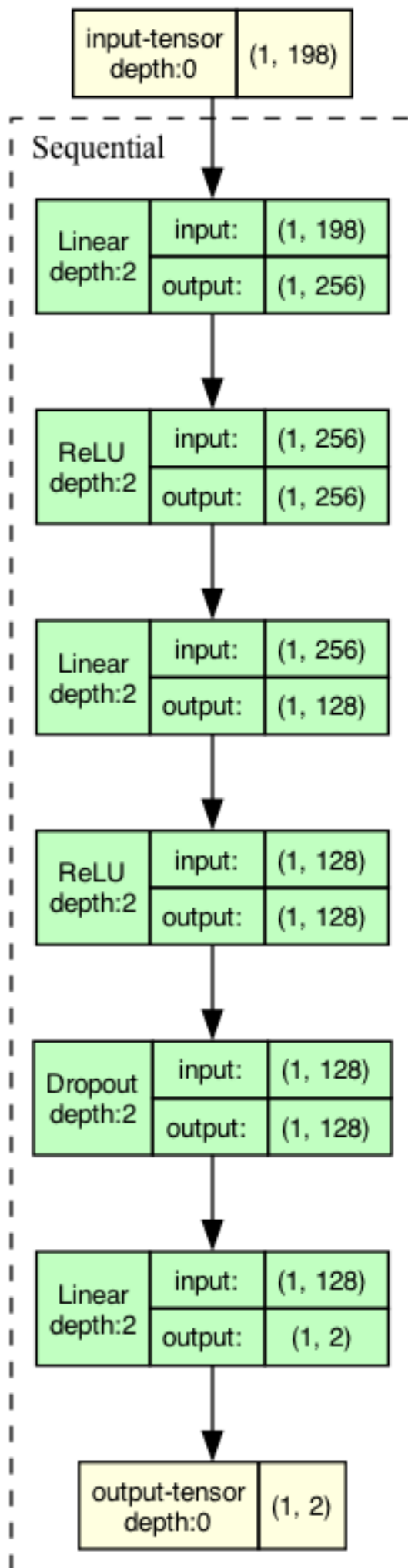


Figure 12: Phase 1 PCA — Best MLP architecture.

3.2 Phase 2

Explained Variance vs Accuracy — SVM & MLP Variants (Hidden sizes:128, 64, 32)

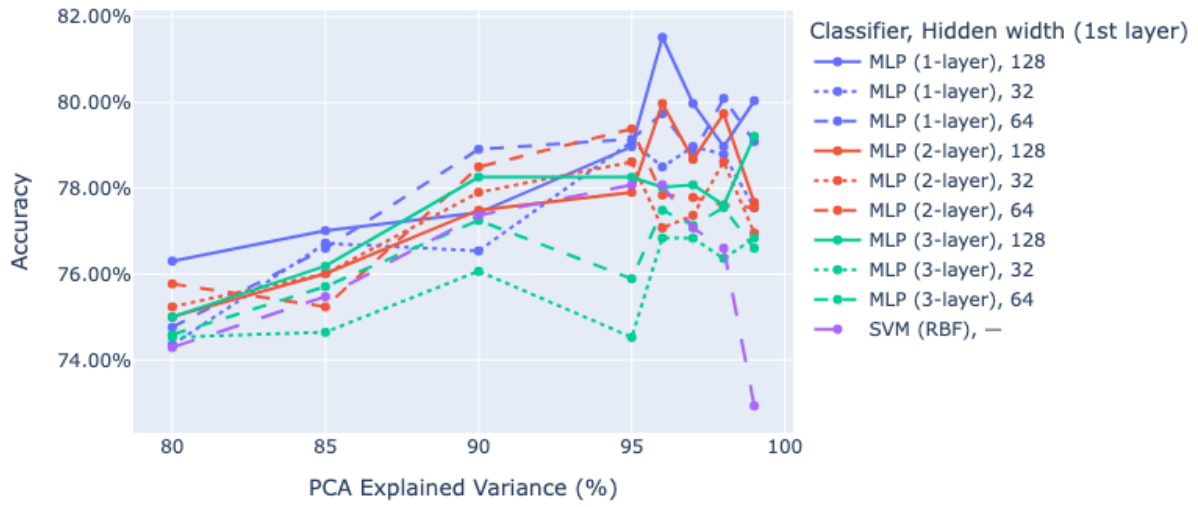


Figure 13: Phase 2 — PCA explained variance vs model accuracy (multiple models).

Best Model and Feature Combination:

96% explained variance, MLP with 1 hidden layer (128), dropout 0.2.

Table 2: Phase 2 PCA — Best model classification report.

	precision	recall	f1-score	support
drummy (0)	0.78	0.86	0.82	926
tight (1)	0.81	0.71	0.75	762
accuracy		0.79		1688
macro avg	0.80	0.79	0.79	1688
weighted avg	0.79	0.79	0.79	1688

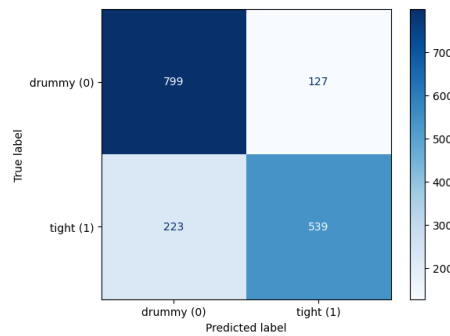


Figure 14: Phase 2 PCA — Best model confusion matrix.

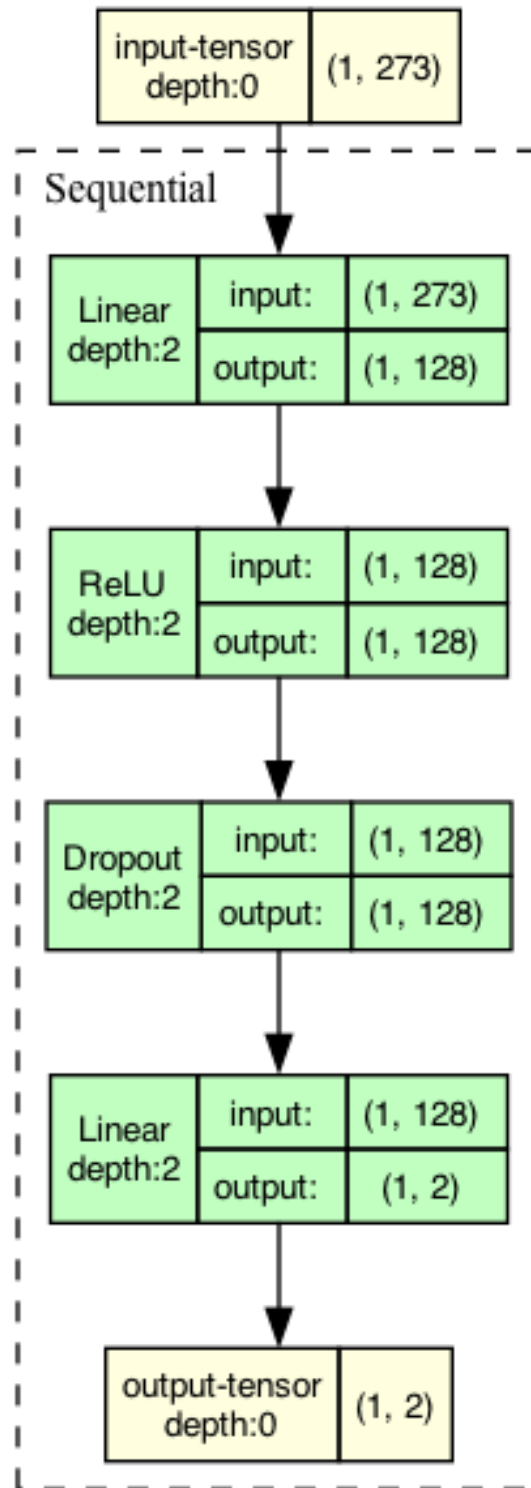


Figure 15: Phase 2 PCA — Best MLP architecture.

4 Deep Learning Models and Generalization Techniques

PCA+MLP provided a strong baseline but cannot exploit the temporal structure of impacts. We therefore explored end-to-end CNNs and MFCC-based hybrids. Split: **90% train / 10% test**.

Generalization:

- **Tanh** after convolutional stages: smoother feature maps and polarity preservation.
- **BatchNorm1d** on early residual outputs: stabilized feature statistics.
- **Dropout 0.6** in classifier heads: stronger regularization.
- **Label smoothing** $\epsilon = 0.1$: mitigated overconfidence.

4.1 Phase 1 — ResNet (End-to-End)

Hyperparameters: Epochs = 20, Batch size = 32, LR = 10^{-3} , Dropout = 0.6.

Table 3: Phase 1 ResNet — Classification report.

	precision	recall	f1-score	support
drummy (0)	1.00	0.98	0.99	221
tight (1)	0.96	0.99	0.97	88
accuracy		0.98		309
macro avg	0.98	0.99	0.98	309
weighted avg	0.98	0.98	0.98	309

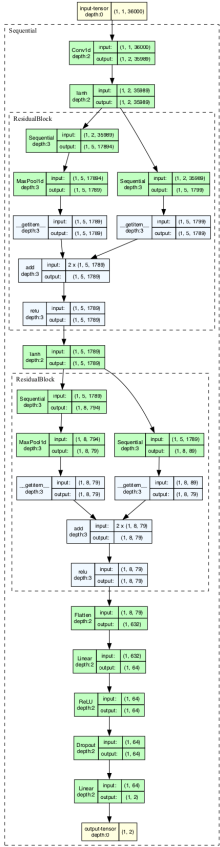


Figure 16: Phase 1 CNN (ResNet-style) model architecture.

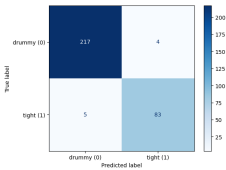


Figure 17: Phase 1 ResNet — Confusion matrix.

4.2 Phase 2 — ResNet (End-to-End)

Hyperparameters: Epochs = 20, Batch size = 90, LR = 10^{-3} , Dropout = 0.6, Weight decay = 10^{-3} .

Table 4: Phase 2 ResNet — Classification report.

	precision	recall	f1-score	support
drummy (0)	0.87	0.92	0.90	466
tight (1)	0.89	0.83	0.86	363
accuracy		0.88		829
macro avg	0.88	0.88	0.88	829
weighted avg	0.88	0.88	0.88	829

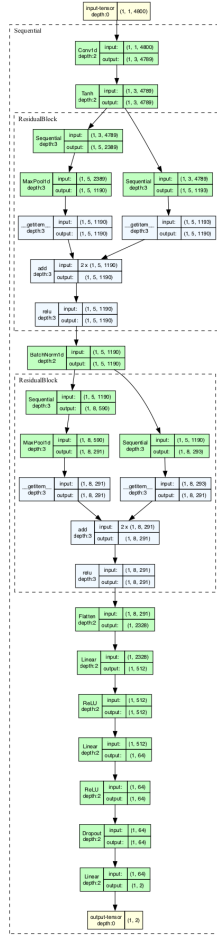


Figure 18: Phase 2 CNN (ResNet-style) model architecture.

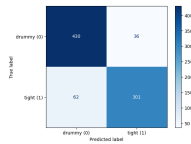


Figure 19: Phase 2 ResNet — Confusion matrix.

4.3 Phase 2 — MFCC + MLP

MFCC config: $n_mfcc = 40$, $n_fft = 1024$, $hop_length = 512$, $n_mels = 40$. Input waveform shape $[1, 4800]$.

Table 5: Phase 2 MFCC + MLP — Classification report.

	precision	recall	f1-score	support
drummy (0)	0.91	0.90	0.91	466
tight (1)	0.88	0.89	0.88	363
accuracy		0.90		829
macro avg	0.89	0.90	0.89	829
weighted avg	0.90	0.90	0.90	829

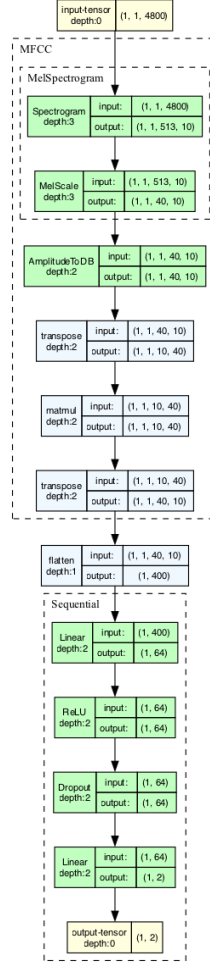


Figure 20: Phase 2 MFCC + MLP — Model diagram.

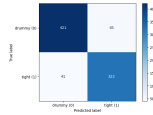


Figure 21: Phase 2 MFCC + MLP — Confusion matrix.

4.4 Phase 2 — MFCC + ResNet

Input: MFCC feature maps; **Backbone:** ResNet-style CNN.

Table 6: Phase 2 MFCC + ResNet — Classification report.

	precision	recall	f1-score	support
drummy (0)	0.94	0.93	0.93	466
tight (1)	0.91	0.92	0.92	363
accuracy		0.93		829
macro avg	0.92	0.93	0.93	829
weighted avg	0.93	0.93	0.93	829

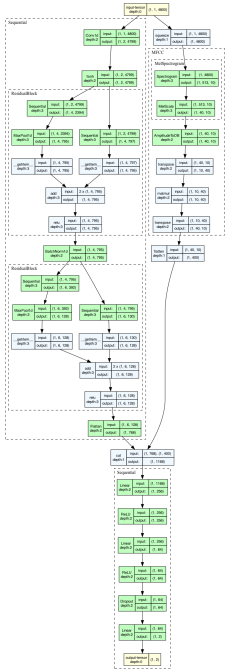


Figure 22: Phase 2 MFCC + ResNet — Model diagram.

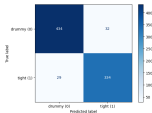


Figure 23: Phase 2 MFCC + ResNet — Confusion matrix.

5 Model Discussion

Phase 1. The end-to-end ResNet-style model reached **98% accuracy**, outperforming PCA+MLP and SVM baselines.

Phase 2. The **MFCC + ResNet** model achieved the best overall balance of precision/recall across both classes (0.93 accuracy), closely followed by **MFCC + MLP** (0.90). Pure waveform ResNet (0.88) also performed strongly.

Generalization effects:

- **Label smoothing** contributed roughly $\sim +2\%$ accuracy.
- **Tanh** after convolutions added $\sim +2\%$, likely by preserving negative polarity in oscillatory signals.

6 Future Work

- Address the discrepancy in input size (Phase 1: 0.75 s / 36k samples; Phase 2: 0.1 s / 4.8k) without resampling—e.g., hierarchical or multi-resolution front ends that accept variable-length inputs.
- Design a **unified architecture** that can process both datasets directly.
- Implement a **robust preprocessing** pipeline to automatically detect & remove mishits (statistical anomaly detection or similarity search seeded by curated mishits).

7 Conclusion

Across multiple approaches, **MFCC + ResNet** delivered the strongest Phase 2 performance with balanced class recall, while an end-to-end **ResNet** saturated Phase 1 near 98% accuracy. Small architectural choices (**label smoothing**, **tanh**) consistently added measurable gains. Future work will target unified input handling and automated data-quality filtering.